

GUIDELINES

1. Don't Try to Build Everything

> This is the biggest hackathon killer.

> Your problem statement is massive.

You CANNOT perfectly solve:

- Heat mapping
- Climate simulation
- AI prediction
- City planning
- Cost optimization
- Real-time satellite integration

in limited time.

Instead:

> ****Build ONE strong core**

Recommended winning core:

“AI-driven hotspot detection + optimized cooling recommendation”

Then add small premium features.

****This wins more than 20 unfinished features.**

2. Your Biggest Weapon = Storytelling

Judges remember stories.

Your pitch should feel like:

“Pune is becoming hotter yearly. Authorities struggle to identify high-risk heat zones and decide the best cooling intervention. Team Orion developed an explainable geospatial AI system that identifies hotspots and recommends optimized cooling strategies using satellite and meteorological data.”

That sounds premium.

Never say:

“We trained a random forest model.”

No judge cares initially.

First tell impact. > Then tech.

3. Choose One City and Dominate It

Huge advice.

DO NOT:  Train on all India

DO:  One city deeply

Example:

> **Pune

Why?

- Data easier
- Faster experimentation
- Local relevance
- Easier validation

You can say:

“Framework is scalable to other Indian cities.”

****Judges will accept.**

4. Documentation = Secret Weapon

>> Most teams ignore documentation.

You should create:

Research Bible (Master Doc)

Contains:

- Problem statement breakdown
- Papers
- Dataset links
- Model choices
- Experiments
- Failures
- Results
- Screenshots
- Meeting notes

Why?

>> Because during PPT:

Someone forgets? — — No issue.

Everything documented.

5. DAILY Internal Review

Every 3 days ask:

- 1. What is done?**
- 2. What is blocked?**
- 3. What failed?**
- 4. What changed?**

This prevents chaos.

7. GitHub Discipline **(CRITICAL)**

Every member must push code.

Structure:

- main
- development
- frontend
- ml-model

Use commits like:

- Added hotspot prediction model
- Integrated NDVI pipeline
- Fixed GIS preprocessing

Looks professional.

Judges sometimes check.

8. Start Presentation Early

****Huge mistake:**

>> Teams make PPT one night before.

Winning teams:

Build PPT alongside project.

Every week:

Add:

- Results
- Graphs
- Screenshots

****Final PPT becomes easy.**

9. Create a WOW Factor

You need one feature that makes judges say:

“Oh wow.”

Examples:

1. Option A (Recommended)

Interactive map:

User clicks region →

AI says:

1.Heat Cause

- Low vegetation
- High concrete density

2.Best Fix

- Add 20% tree cover

3.Expected Cooling

- 2.3°C reduction

Very impressive.

2. Option B

Future prediction:

“How city will look in 2030 heat-wise.”

3. Option C

Cost vs Cooling graph.

Method	Cost	Cooling
Trees	Low	High
Cool roof	Medium	High

****Looks practical.**

10. Prepare Answers for Judge Questions

You WILL be grilled.

Prepare beforehand.

1. Why Random Forest?

> Because interpretable and robust for tabular geospatial data.

2. Why not Deep Learning?

> Insufficient labeled spatial thermal data + interpretability concerns.

3. Why these datasets?

> Satellite + meteorological complement each other.

4. Limitations?

> Cloud cover, temporal variation, city-specific training.

****Judges LOVE honesty.**

11. Time Allocation Rule

Spend:

- 40% Model
- 20% Dashboard
- 20% Documentation
- 20% Presentation

Most teams do:

>> 90% coding.

****Then lose.**

12. Don't Fake Results

****Very important.**

Judges are experienced.

Instead of:

>>  "99.9% accuracy"

Say:

>>  "Model achieved robust predictive performance with interpretable outputs."

**** Be realistic.**

13. Use Research Words Carefully

These words sound strong:

- Explainable AI
- Geospatial Intelligence
- Physics-informed ML
- Urban Heat Island Analysis
- Scenario Optimization
- Decision Support System
- Sustainable Urban Planning

Use in PPT.

****But understand them.**

14. One Team Rule

****No ego.**

No:

“This is my work.”

Everything shared.

All 3 must know:

- Problem
- Model
- PPT
- Demo

Judges ask random member questions.

**** If one member fails badly → huge damage.**

15. My Biggest Advice

Don't chase:

“Most advanced project”

Chase:

Most complete project

Winning project checklist:

- ✓ Works
- ✓ Clean architecture
- ✓ Good reasoning
- ✓ Explainable
- ✓ Real use case
- ✓ Strong demo
- ✓ Strong speaking

strongest chance is:

“Explainable Geospatial AI + Cooling Optimization Dashboard for Pune”

PROGRESS

1. We have to make a Progress report of everyone work on our own

2. Daily meet online :

1. What did we complete today?
2. What is blocked?
3. What will we finish tomorrow?

3. Do have everyone's report with them

